👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.1

# Division Expressions with Fractions and Whole Numbers

Lesson 6-1 · Teacher Copy

**Name:** _____ **Date:** _____ **Class:** _____

EVIDENCE LAB MISSION

## The Fraction Vault

You are the lead forensic analyst at the Reyes Detective Agency. A 3-foot strip of evidence tape must be cut into exact 1/4-foot sections before it can be logged — and the vault that holds the case file only opens once you know how many pieces that makes. Interpret fraction division and crack the case.

## My Notes

**Level 3 Enrichment**

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can explain what dividing with a whole number and a unit fraction means — how many groups fit, or how much is in one group — find the quotient with a model, and check it by multiplying.**

**Language Objective: I can explain why dividing by a unit fraction gives the same answer as multiplying by its reciprocal, using the words reciprocal, divisor, and quotient.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

**WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

**Reciprocal** | **Quotient** | **Dividend** | **Divisor** | **Unit fraction** | **Multiplicative inverse** | **Tape diagram**

*Level 3: try the blanks from memory first, then check the bank.*

1. Flipping the numerator and denominator gives the _____ .
2. The answer to a division problem is the _____ .
3. In 6 ÷ 1/2, the number 6 being divided is the _____ .
4. In 6 ÷ 1/2, the number 1/2 we divide by is the _____ .
5. A fraction with a numerator of 1, like 1/5, is a _____ .
6. Two numbers whose product is 1 are _____ of each other.
7. A bar cut into equal parts to show sharing is a _____ .

## Write About the Math

**How does interpreting fraction division help the detective?**
*¿Cómo ayuda interpretar la división de fracciones al detective?*

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Reciprocal** — *Recíproco*
- ☐ **Quotient** — *Cociente*
- ☐ **Dividend** — *Dividendo*
- ☐ **Divisor** — *Divisor*
- ☐ **Unit fraction** — *Fracción unitaria*

**Model:** *There will be more than 3 sections because each piece is smaller than a whole foot.* — Students reason that dividing by a number less than 1 gives a quotient greater than the dividend, so there are more than 3 sections.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The detective can mark ___ sections because 6 ÷ 1/3 = ___.**
- **My answer is ___ because ___.**
  *Mi respuesta es ___ porque ___.*

### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ___.**
  *Mi afirmación es ___.*
- **My evidence is ___, so ___.**
  *Mi evidencia es ___, entonces ___.*

**Explain**

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

## Answer Key & Teacher Guide

1. **Notes 1:** Reciprocal
2. **Notes 2:** Quotient
3. **Notes 3:** Dividend
4. **Notes 4:** Divisor
5. **Notes 5:** Unit fraction
6. **Notes 6:** Multiplicative inverse
7. **Notes 7:** Tape diagram

8. **Watch for this mistake:** *A common mistake in Interpret Division of Fractions is assuming that dividing always makes the answer smaller. For 4 ÷ 1/3, students reason 'dividing shrinks the number' and compute 4 ÷ 3 ≈ 1.3 instead of asking how many 1/3-foot pieces fit into 4 feet (each whole foot holds 3 thirds, so 4 × 3 = 12). Before you submit, check: if the divisor is a fraction smaller than 1, the quotient must be BIGGER than the dividend, not smaller.*
9. **Try It 1:** B. 2/3 ÷ 2 — *Splitting 2/3 into 2 equal pieces is 2/3 ÷ 2. Multiply by the reciprocal of the divisor: 2/3 × 1/2 = 2/6 = 1/3 yard.*
10. **Try It 2:** — *3/4 ÷ 1/4 asks how many one-fourth pieces make up three-fourths. The fraction bar shows 3/4 already built out of fourth-size pieces, so counting those pieces gives the quotient: 3. Multiplying by the reciprocal agrees — 3/4 × 4/1 = 12/4 = 3. Neither number is a whole number here, so there is nothing to write over 1.*